



Mathematical Statistics Week 6

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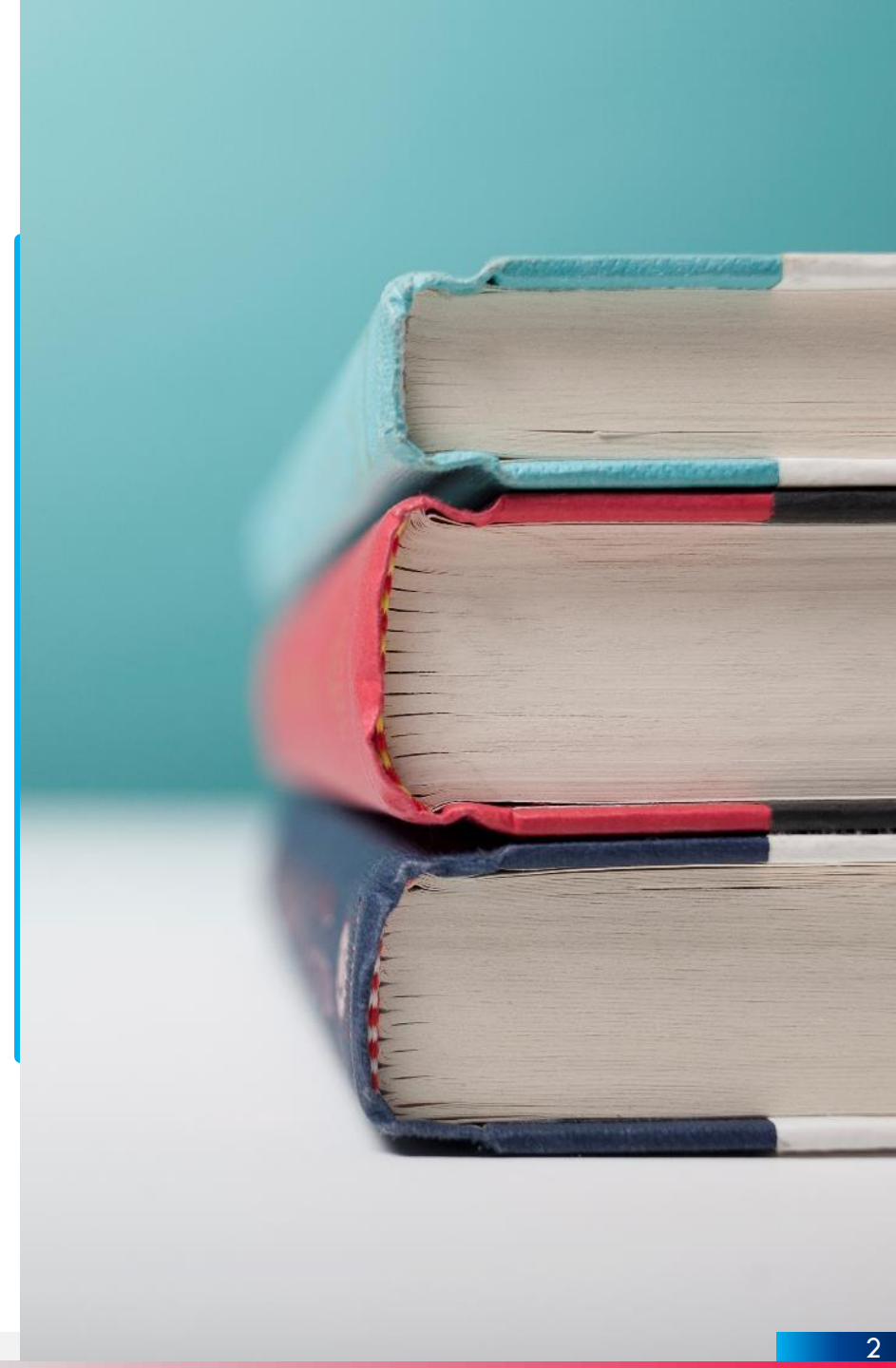
Mathematics and Natural Sciences
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Summary & Outline

This week, we will delve into hypothesis testing, inferences on proportions, and comparing two means. These concepts are fundamental in statistical analysis and will help you understand how to make inferences about populations based on sample data.

- Hypothesis Testing: Understanding the process of testing claims about population parameters.
- Inferences on Proportions: Estimating and testing proportions in populations.
- Comparing Two Means: Techniques for comparing means from two different populations.
- Summary: Recap of key concepts and methods.



Tests Concerning a Single Mean

- Tests Concerning a Single Mean:

- Hypotheses:

- Null hypothesis (H_0): $\mu = \mu_0$
- Alternative hypothesis (H_1): $\mu \neq \mu_0$, $\mu > \mu_0$, or $\mu < \mu_0$

- Test Statistic:

- $$Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

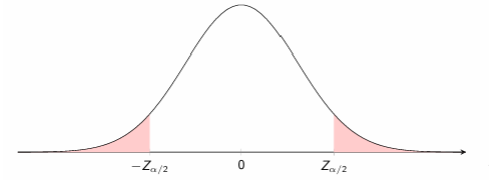
- Critical Regions:

- Two-tailed: $|Z| > Z_{\alpha/2}$
- Right-tailed: $Z > Z_{\alpha}$
- Left-tailed: $Z < -Z_{\alpha}$

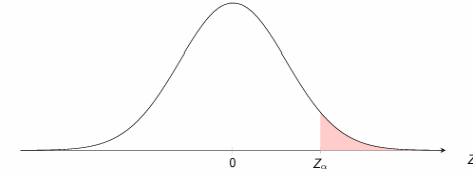
- Example:

- Given a sample mean, population standard deviation, and sample size, calculate the test statistic and compare it to the critical value to make a decision.

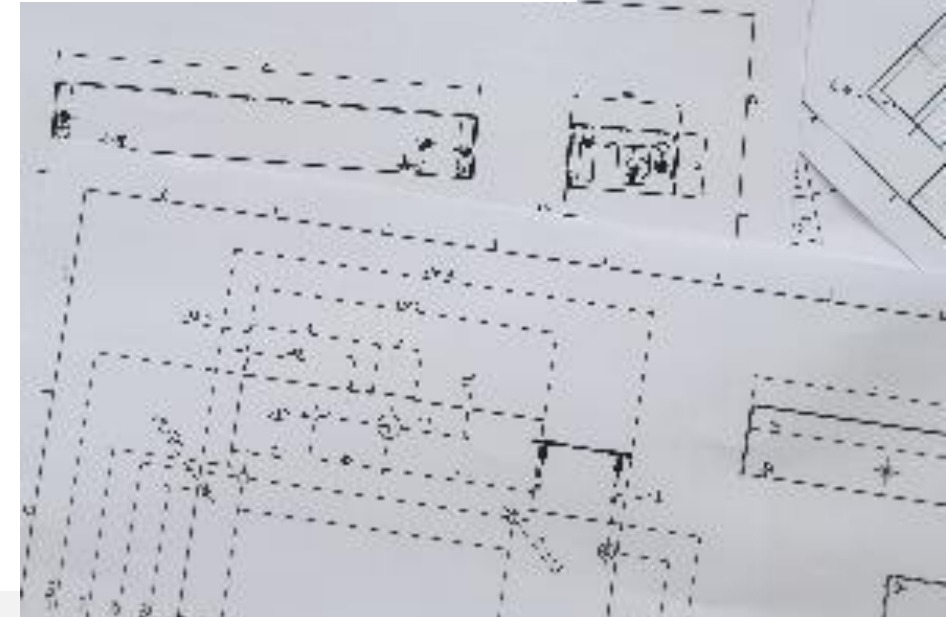
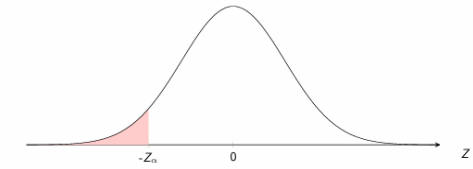
$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > Z_{\alpha/2} \quad \text{or} \quad Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < -Z_{\alpha/2}$$



$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > Z_{\alpha}$$



$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} < -Z_{\alpha}$$



Example #1

Problem:

A random sample of 100 recorded deaths in the United States showed an average life span of 71.8 years. Assuming a population standard deviation of 8.9 years, test if the mean life span is greater than 70 years at a 0.05 significance level.

• Solution:

- Hypotheses: $H_0: \mu=70$, $H_1: \mu>70$
- Significance level: $\alpha=0.05$
- Critical region: $Z>1.645$
- Test statistic: $Z = \frac{71.8-70}{\frac{8.9}{\sqrt{100}}} = 2.02$
- Decision: Reject H_0 and conclude that the mean life span is greater than 70 years.



Tests on a Single Mean (Variance Unknown)

Tests on a Single Mean (Variance Unknown):

- Hypotheses:
 - Null hypothesis (H_0): $\mu = \mu_0$
 - Alternative hypothesis (H_1): $\mu \neq \mu_0$, $\mu > \mu_0$, or $\mu < \mu_0$
- Test Statistic:
 - $t = \frac{\bar{X} - \mu_0}{\frac{s}{\sqrt{n}}}$
- Critical Regions:
 - Two-tailed: $|t| > t_{\alpha/2, n-1}$
 - Right-tailed: $t > t_{\alpha, n-1}$
 - Left-tailed: $t < -t_{\alpha, n-1}$
- Example:
 - Given a sample mean, sample standard deviation, and sample size, calculate the test statistic and compare it to the critical value to make a decision.

Example #2

Problem:

An electric institute has published figures on the number of kilowatt hours used annually by various home appliances. It is claimed that a vacuum cleaner uses an average of 46 kilowatt hours per year. If a random sample of 12 homes included in a planned study indicates that vacuum cleaners use an average of 42 kilowatt hours per year with a standard deviation of 11.9 kilowatt hours, does this suggest at the 0.05 level of significance that vacuum cleaners use, on average, less than 46 kilowatt hours annually? Assume the population of kilowatt hours to be normal.

• Solution:

- Hypotheses: $H_0: \mu = 46$ kilowatt hours, $H_1: \mu < 46$ kilowatt hours
- Significance level: $\alpha=0.05$
- Critical region: $t < -1.796$
- Test statistic: $t = \frac{42-46}{\frac{11.9}{\sqrt{12}}} = -1.16$
- Decision: Do not reject H_0 and conclude that the average number of kilowatt hours used annually by home vacuum cleaners is not significantly less than 46.



Relationship to confidence interval

- Relationship to Confidence Interval Estimation:
 - Concept:
 - Hypothesis testing and confidence interval estimation are closely related. Both methods use the same underlying principles and formulas.
 - Confidence Interval:
 - For a single mean with known variance: $\bar{X} \pm Z_{\frac{\alpha}{2}} \sigma / \sqrt{n}$
 - For a single mean with unknown variance: $\bar{X} \pm t_{\frac{\alpha}{2}, n-1} s / \sqrt{n}$
 - Decision Rule:
 - If the hypothesized value μ_0 lies outside the confidence interval, reject H_0 .



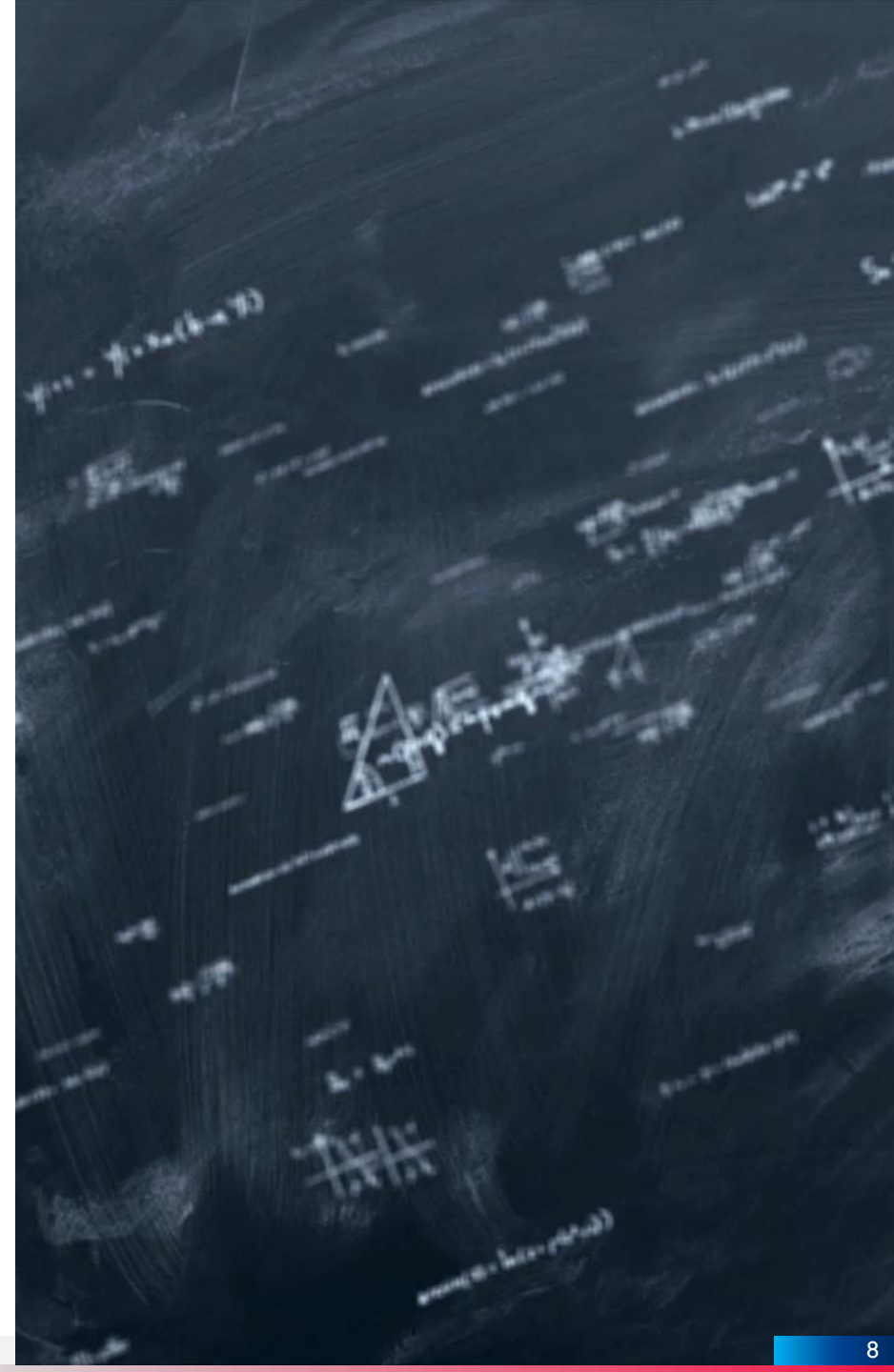
Inferences on Proportions

- Definition:

- Proportion ($\hat{p}$): $\hat{p} = \frac{X}{n}$ where X is the number of successes and n is the sample size.

- Confidence Interval: $\hat{p} \pm Z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

- Sample Size Calculation: $n = \frac{Z_{\frac{\alpha}{2}}^2 \hat{p}(1-\hat{p})}{d^2}$ where d is the desired margin of error.



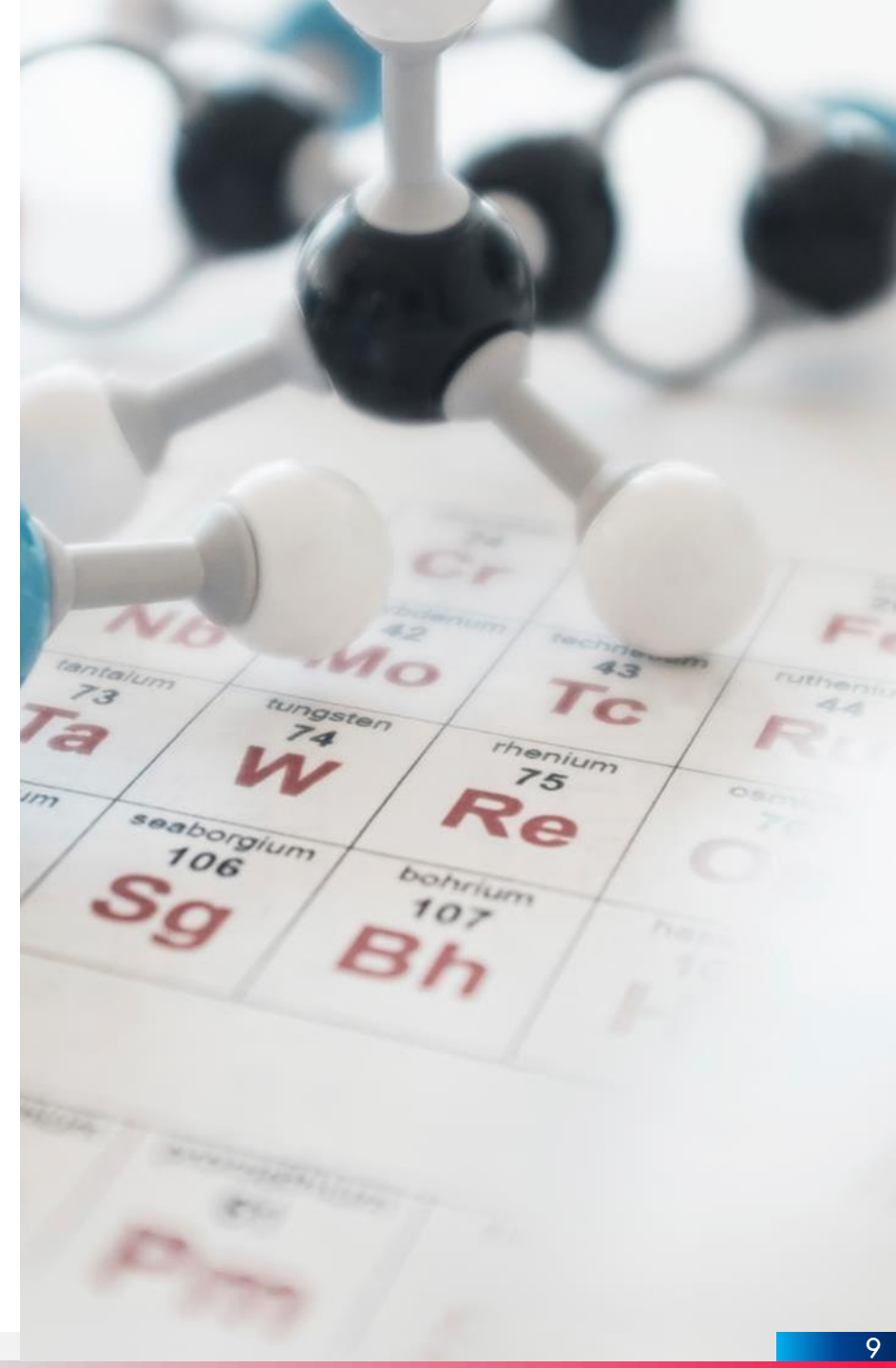
Example #3 & #4

To estimate the reliability of 4 GB dynamic RAMs, a sample of 100 is tested. 91 out of 100 circuits performed properly. What is the point estimate for p ?

- Solution:
 - $\hat{p} = \frac{91}{100} = 0.91$
 - Define the 95% confidence interval on p for the previous example.
 - $\hat{p} = 0.91, n = 100, Z_{\frac{\alpha}{2}} = 1.96$
 - Confidence interval: $0.91 \pm 1.96 \sqrt{\frac{0.91 \cdot 0.09}{100}} = 0.91 \pm 0.056$

How large a sample is required to estimate the proportion of 4GB dynamic RAMs that operate properly to within 1% point with 95% confidence?

- $\hat{p} = 0.91, d = 0.01, Z_{\frac{\alpha}{2}} = 1.96$
- $n = \frac{1.96^2 \cdot 0.91 \cdot 0.09}{0.01^2} = 3147$



Inference on Proportions: Testing Hypotheses on p (Overview)

- Test statistic

- $$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

- Null and alternative

- Right-tailed: $H_0 : p = p_0, H_1 : p > p_0$
 - Left-tailed: $H_0 : p = p_0, H_1 : p < p_0$
 - Two-tailed: $H_0 : p = p_0, H_1 : p \neq p_0$

- Rejection regions

- Right: $Z > Z_\alpha$
 - Left: $Z < -Z_\alpha$
 - Two-tailed: $|Z| > Z_{\frac{\alpha}{2}}$

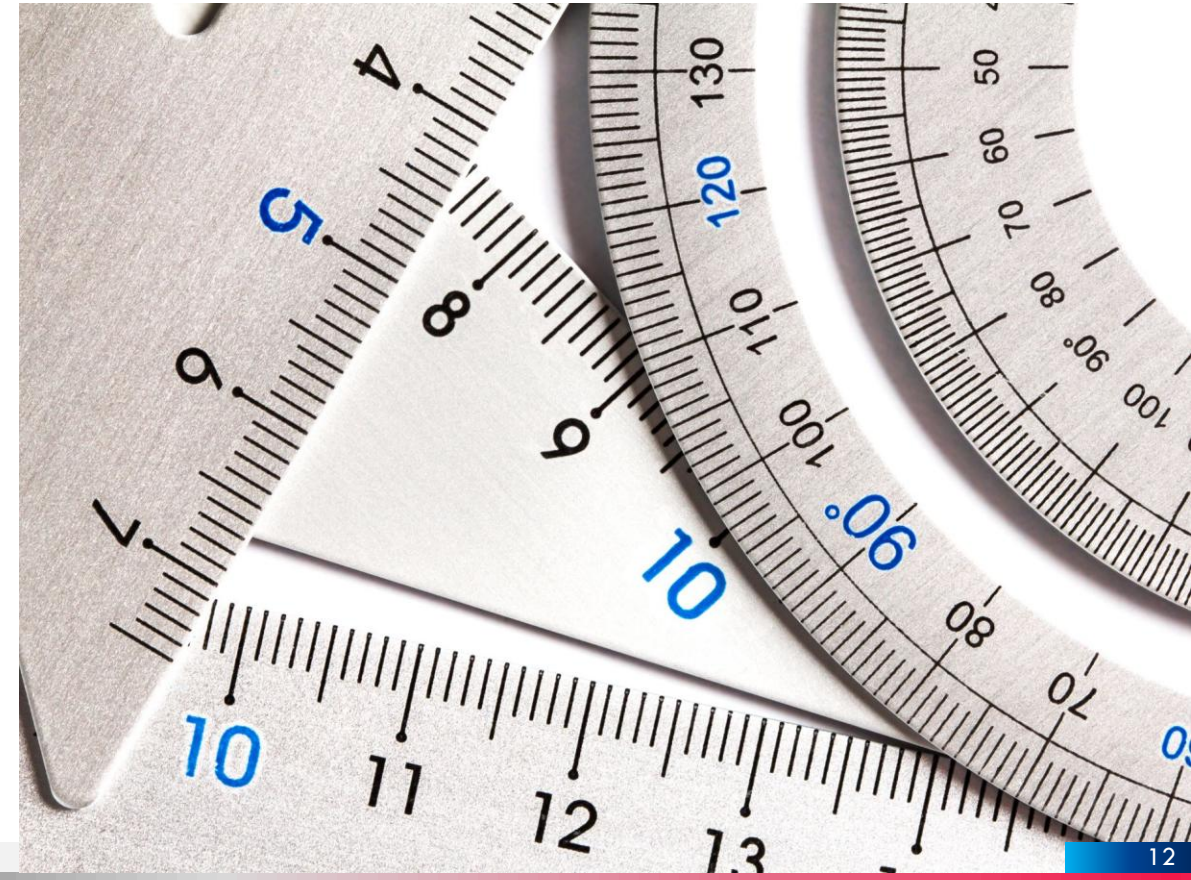
Inference on Proportions: Testing Hypotheses on p (Critical Values & p-Value)”

- The possible outcomes of the test hypothesis are...
 - For right-tailed test: H_0 is rejected in favour of H_1 if the observed value of the test statistics is a large positive number;
 - For left-tailed test: large negative numbers lead to rejection;
 - For two-tailed test: H_0 is rejected for values of the test statistic that are too large in either the positive or negative sense.
- Critical-value approach
 - Compare computed Z to Z_α or $Z_{\frac{\alpha}{2}}$ as on previous slide.
- p-Value approach
 - $$\text{p-value} = \begin{cases} 1 - \Phi(Z) & \text{right-tailed} \\ \Phi(Z) & \text{left-tailed} \\ 2[1 - \Phi(|Z|)] & \text{two-tailed} \end{cases}$$
 - Decision rule: Reject H_0 if $\text{p-value} < \alpha$.
- small numeric example $n=200$, $\hat{p}=0.755$, test $p_0=0.7$, yields $Z=1.697$, $\text{p-value} \approx 0.045 \rightarrow \text{reject at } \alpha=0.05$.)



Comparing Two Means & Two Variances (Overview)

- Independent samples from two normal
 - Means μ_1, μ_2 ; variances σ_1^2, σ_2^2 .
 - Point estimator $\bar{X}_1 - \bar{X}_2$
 - Var of difference $\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$
- Three cases
 - Both known $\rightarrow$ Z-test
 - Unknown, equal variances $\rightarrow$ pooled t -test
 - Unknown, unequal variances $\rightarrow$ Welch's t -test



Comparing Two Means: σ_1, σ_2 Known

- Confidence interval

- $(\bar{X}_1 - \bar{X}_2) \pm Z_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

- Hypothesis test

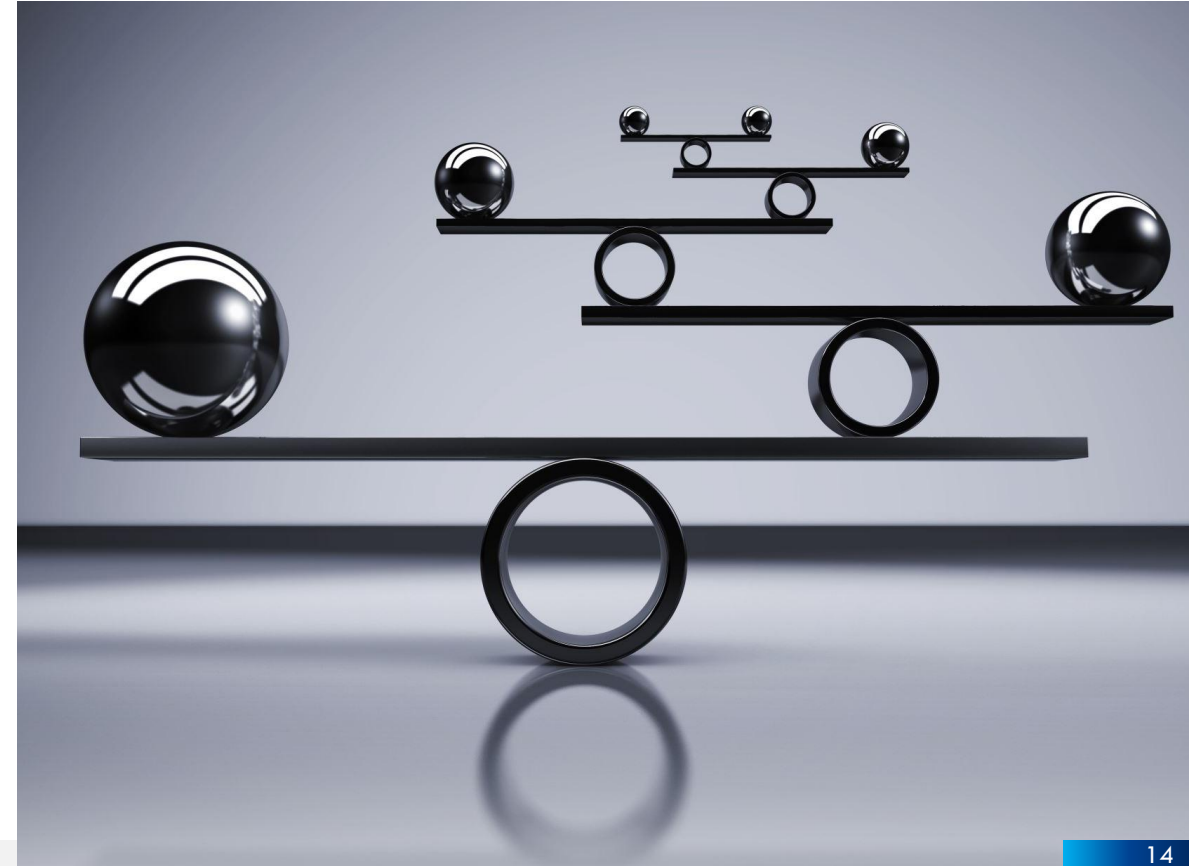
- $Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$

- Reject $H_0: \mu_1 - \mu_2 = d_0$ if $|Z| > Z_{\frac{\alpha}{2}}$ (two-tailed)

- (Variance-known two-sample Z-procedure.)

Comparing Two Means: Equal Variances Unknown

- Pooled variance
 - $S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$
- Confidence Interval and Test
 - $(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
 - $t = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$
- Reject $H_0: |t| > t_{\frac{\alpha}{2}, n_1+n_2-2}$ (two-tailed)
- *(Student's two-sample ttt-test under equal-variances.) lec6*



Comparing Two Means: Unequal Variances Unknown

- Welch's t-test

- $$t = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- Approximation to degrees of freedom (Welch–Satterthwaite)

- $$\nu \approx \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$$

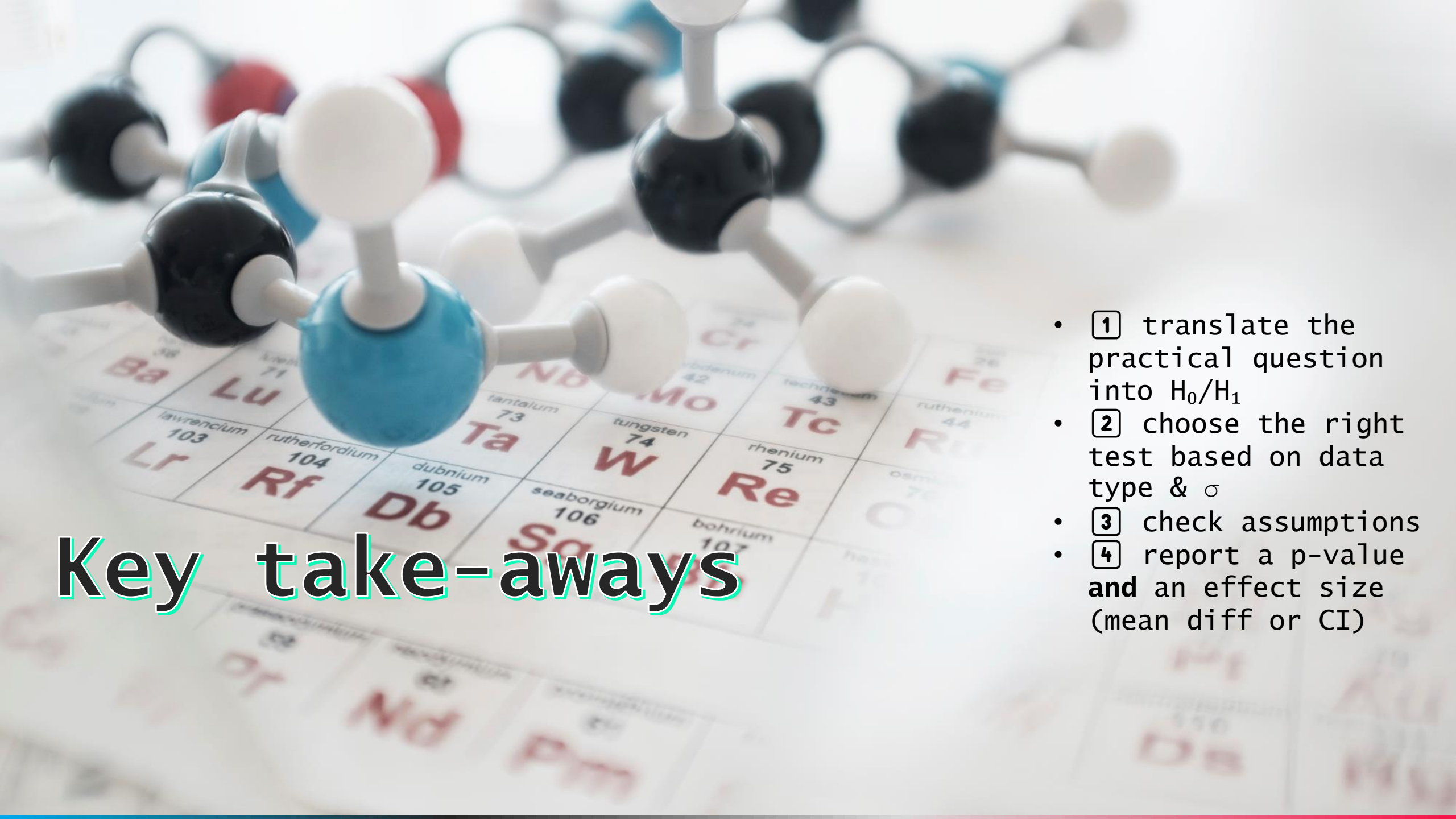
- **Decision**

- Compare $|t|$ to $t_{\frac{\alpha}{2}, \nu}$.

Comparing Two Means: Paired Data

- **When to use:** measurements in “pairs” on the same unit (before/after, matched).
- **Difference:** $D_i = X_{1i} - X_{2i}$, $i = 1, \dots, n$.
- One-sample t-test on differences
 - $\bar{D} \pm t_{\frac{\alpha}{2}, n-1} S_D / \sqrt{n}$ and $t = \frac{\bar{D} - d_0}{\frac{S_D}{\sqrt{n}}}$
- Degrees of freedom = $n-1$
- Reject $H_0: \mu_D = d_0$ if $|t| > t_{\frac{\alpha}{2}, n-1}$
- Reduces to a one-sample problem on the differences.





Key take-aways

- ① translate the practical question into H_0/H_1
- ② choose the right test based on data type & σ
- ③ check assumptions
- ④ report a p-value and an effect size (mean diff or CI)